HACKATHON BUILDER PACKET

KTx-CONNECT (aka DEAP)

Kidney Tx Communication & Online Nudges for Nonadherence Early Capture & Tracking

1 The challenge

Design a way to run the MAPS loop (find a patient's medication barriers, agree on one plan, check back) between clinic visits and at centers without a dedicated adherence team.

WHY IT MATTERS

Teens and young adults with a kidney transplant must take anti-rejection medicine every day for as long as the kidney lasts. About 40% of them struggle with this, and their transplants fail at roughly 3 times the rate of other age groups. A failed transplant usually means a return to dialysis.

WHAT ALREADY WORKS

Cincinnati Children's built MAPS (Medication Adherence Promotion System) as standard care in 2015–2016. Afterward, late acute rejection fell by about half: 1.61 to 0.88 episodes per 100 patient-months. A cost model found MAPS saved about $9,106 per patient and added 0.32 quality-adjusted life years.

THE PROBLEM TO SOLVE

MAPS runs on people: trained staff, a paper or tablet checklist entered into the chart, and a team member who calls back within 2 weeks. Most transplant centers cannot staff that. Most missed doses also happen at home, between visits, when nobody on the care team can see them.

DESIGN QUESTION

How might we keep MAPS's barrier → plan → follow-up loop running between visits and at any center, while keeping a human in the conversations that need one?

Any form factor is fair game: app, text messages, clinic kiosk, inside the electronic health record, a wearable, a pharmacy touchpoint, or something we haven't thought of.

WANT TO SEE HOW WE PICTURED IT?

Our narrated concept demo is at demo.graftwisehealth.com. It shows one take: a clinician portal plus patient and caregiver web apps. We are not committed to that form factor. Use it to understand the clinical loop, not as a template to copy.

2 Who the users are

There are three sides: the patient, the caregiver and the transplant team. A design that serves only one of them will stall. The people below are composites drawn from common patterns, not real patients.

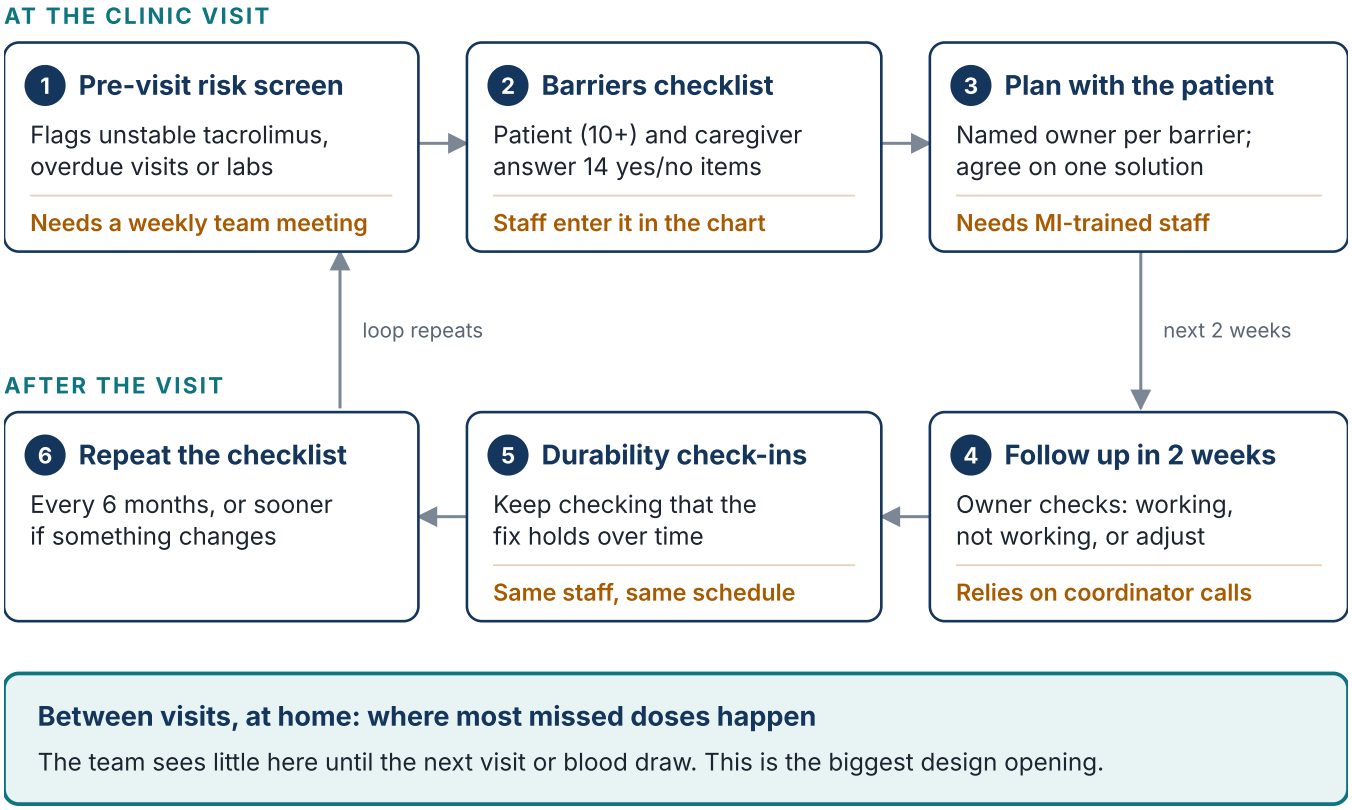
PERSONA	WHAT THEY WANT	WHAT GETS IN THE WAY	WHAT IT MEANS FOR YOUR DESIGN
Maya, 15 Transplant at age 9, in high school	To feel normal; not to be nagged	Forgets on weekends and at sleepovers; doesn't want friends to see her take pills	Discreet and phone-native; must not feel like parental surveillance
Jordan, 20 In college, moving to adult care	Independence; less hovering from home	Runs out of medicine; refills and insurance used to be Mom's job; irregular sleep schedule	Supports the hand-off of responsibility from parent to patient
Leo's mom Leo is 8; she completes the checklist for him	To get it right while juggling work and siblings	Leo hates the taste and can't swallow pills; cost; may read English as a second language	The caregiver is the user; low literacy; Spanish and Arabic matter
Transplant coordinator Nurse who owns most action plans	To know which patients need her this week	Follow-up calls slip when clinic is busy; she gets too many alerts already	A short to-do list she trusts, not another alert stream
Social worker and psychologist Receive referrals	A clean handoff with context	Referrals arrive without the "why"	Routing carries the barrier, the history and who already tried what

Two things to keep in mind

- **Patients 10 and older and their caregivers each answer separately.** When they disagree, that tells the team something about the family.
- **Teens are entitled to some privacy from their parents.** Decide what each person can see.

3 How MAPS works in clinic today

MAPS is a loop, not a single screen. Each step works because a specific staff member does it, which is why other centers have trouble copying it.



Between visits, at home: where most missed doses happen

The team sees little here until the next visit or blood draw. This is the biggest design opening.

Where the step depends on dedicated staff Where today's loop has no visibility

The MAPS loop today · 6 steps, repeating every 6 months

The colored lines show where staff time goes. The band at the bottom is where most missed doses happen. A strong design either takes work out of the colored lines or adds visibility to the band, ideally both.

4 What must carry over, and what's open

The six functions below are where we believe MAPS gets its effect. Keep each of them, but feel free to reinvent how each one is delivered.

MUST KEEP	WHAT IT MEANS IN PRACTICE
1. Ask about barriers, regularly and in a structured way	The 14-item Barriers Assessment Tool (BAT): "What gets in the way of you taking your medicine?" Today it's done every 6 months, or sooner if something changes.
2. Hear from the patient and the caregiver separately	Caregiver only for patients under 10; both for patients 10 and older. Disagreement between them is a signal.
3. Give every barrier a named owner	Each barrier routes to a specific role (see the reference below). Clear ownership is part of why MAPS worked.
4. Agree on one concrete plan per barrier	One solution to try, with who, what, how, where and when. It's made *with* the patient, not handed to them.
5. Close the loop	The owner checks back within 2 weeks: working, not working, or adjust. Then keep checking over time (the durability phase).
6. Use risk signals already in the chart	Flags such as tacrolimus levels that swing more than 2 SD across 4 draws, no clinic visit in over 4 months, or no labs in over 6 weeks.

The tone stays nonjudgmental throughout. Missed doses are a problem to solve together, never a failure to punish.

Open for you to redesign

- **Form factor:** where and how the loop happens (app, text, voice, kiosk, the health record, pharmacy, school)
- **Engagement:** why a 15-year-old would open it a third time
- **Shifting responsibility from parent to teen** over several years
- **Triage for staff:** turning between-visit data into a short, trustworthy worklist
- **Self-help for simple barriers** (forgetting, taste) while complex ones still reach a human
- **Access:** language, literacy, disability and families without reliable data plans
- **Medication monitoring:** smart pillboxes have been used, but about 4 in 10 patients offered one in the original study refused it or had technical trouble

One firm limit: motivational interviewing, the counseling technique behind the action-plan conversation, needs a trained person. Your design can prepare that conversation and make it shorter. It should not try to replace it.

5 Reference: the BAT and who owns each barrier

Each of the 14 BAT barriers routes to one response and one owner. This mapping is what lets a busy clinic act on the checklist. The BAT is a yes/no checklist at a 5th-grade reading level and exists in English, Spanish and Arabic. In our manual clinic version this is how we at Cincinnati Children's designated an owner, but this is site specific and can be modified for a new form factor.

#	BARRIER (AS THE PATIENT SEES IT)	RESPONSE	OWNER
1	Forgetting to take them or bring them with you	"Forgetting" action plan	Transplant coordinator
2	Hate the taste	"Taste" action plan	Transplant coordinator
3	Hard to swallow pills	"Pill swallowing" action plan	Transplant coordinator
4	Side effects (e.g., nausea, constipation)	"Side effects" action plan	Transplant coordinator
5	I don't want others to know I take medicine	"Embarrassment" action plan	Transplant coordinator
6	I don't need the medicine	Medication education	Transplant coordinator
7	Too many medicines or too many times a day	Discussion; possibly a simpler regimen	Nephrologist, NP or pharmacist
8	The medicine does not work	Discussion	Nephrologist, NP or pharmacist
9	Difficult to understand medication instructions	Discussion	Nephrologist, NP or pharmacist
10	Ran out of medicine	Referral	Social worker
11	Can't afford the medicine	Referral	Social worker
12	Gets in the way of other activities	Referral	Psychologist
13	Refuse to take them	Referral	Psychologist
14	Inconvenient	Referral	Psychologist

NP = nurse practitioner.

The team also flags three things the checklist doesn't ask about: suspected health-literacy problems (to the social worker), depression, anxiety or ADHD in the patient or parent (to psychology), and organization or executive-function difficulties (to neuropsychological testing).

The five action plans

Shared decision-making handouts made with families. Each ends with one solution to try:

- **Forgetting:** phone or app reminders, or pairing doses with a daily routine such as brushing teeth
- **Taste:** pharmacist flavoring, coatings or syrups, numbing taste buds with a popsicle first, or mixing with food when allowed
- **Pill swallowing:** practicing with candy, changing head position, coating the pill
- **Embarrassment:** carrying and taking medicine discreetly; a short script for explaining it to friends
- **Side effects:** a clinician-facing guide to changing, adding or adjusting medicines

In practice, about 1 in 3 patients reports at least one barrier. The most common are side effects, bad taste and forgetting.

6 What a strong solution looks like

A strong solution keeps the whole loop intact and takes work off the team. A polished screen that drops one of the six must-keep functions is weaker than a rough sketch that keeps all of them.

WE'LL LOOK FOR	STRONG	WEAK
The loop is closed	A barrier found today leads to a plan and a check-back, and someone is accountable for each step	Data is collected, but nobody owns what happens next
Staff time	Fewer calls and less data entry; the coordinator sees a short list of who needs her	A new dashboard and more alerts on top of the current workload
Fits a teen's life	Discreet, quick, worth opening again; feels like help	Nagging, childish or shaming; streaks that punish a missed day
Works for the family	Patient and caregiver each have a role; teen privacy is respected; responsibility can shift over time	One account for the whole family; the parent sees everything
Reaches everyone	5th-grade reading level, multiple languages, works without a new phone or unlimited data	Assumes an iPhone, fluent English and high health literacy
Knows its limits	Self-help for simple barriers; hard ones (cost, mood, refusal) go to a human fast	A chatbot trying to act as the counselor
Measurable	You can say what would show it works: barriers resolved, follow-ups completed, steadier drug levels	"Engagement" as the only result

QUESTIONS WE'LL PROBABLY ASK YOU

1. What does the coordinator actually do with your output on a busy Tuesday afternoon?
2. What happens when a 16-year-old says "I refuse to take them"?
3. What does the parent see, and what does the teen keep private?
4. Why would someone use this in month 6, not just in week 1?
5. How would a center with no psychologist on staff use this?